

Development of a beta-type Stirling heat pump with rhombic drive mechanism by a modified non-ideal adiabatic model

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Summary

In this paper, a thermodynamic model is developed for predicting the performance of a beta-type Stirling heat pump with rhombic drive mechanism for water heater and the model is validated by a 1-kW class prototype Stirling heat pump. In the present model, the working space is divided into expansion space, heat absorber, regenerator, heat rejecter and compression space. The pressure, mass and temperature variations of working fluid in each working space are predicted. The temperature variation of wall boundary is also taken into consideration. The temperature of working fluid in each working space and the temperature of wall boundary are obtained by solving energy equations simultaneously. Eventually, the pressure of working fluid in each working space can be corrected by using empirical formula of pressure drop. All the thermal properties of working fluid and wall boundary in each working space at each time step can be obtained by repeating the above process. Then, the performance of heat pump such as absorbing heat, rejecting heat, indicated power and COP can be calculated. A series of experimental measurements and comparisons are also conducted for validating present model. The results show that the prototype heat pump can produce 904 W heating power and 38°C hot water under 1 LPM water flow rate with 5 bar helium at 1000 rpm.

KEYWORDS

beta type, heat pump, non-ideal adiabatic model, rhombic drive mechanism

1 | INTRODUCTION

With the development of industry and technology, the consumption of energy sources becomes faster and faster, which is often accompanied with environmental pollution and ecological destruction. To solve these problems, a greater number of researchers are investing as much as they can to develop clean energy and to use existing energy more efficiently. For example, heat pump is one of the choices that can use existing energy more efficiently.

Heat pump is a kind of device that can absorb heat from low-temperature reservoir and transport it to

high-temperature reservoir. It is different and more efficient compared to other kind of heaters such as combustion heater or electric heater because the heat generated by heat pump is larger than the work it consumed. This due to the work it consumed is used to pump heat from low-temperature reservoir and then transformed to heat. After that, both of them are transported to high-temperature reservoir together. By considering these reasons, many kinds of heat pump were invented and developed such as adsorption heat pump, chemical heat pump, vapor compression heat pump and steam-driven jet heat pump. According to the type

of heat sources, heat pump can be further classified into water, ground, air and solar source heat pump and they are widely applied in many fields, such as water heater, air-condition, waste heat recovery, drying, evaporation and distillation. Although above heat pumps are able to use existing energy more efficiently, they bring another derivative problem that some of them adopt refrigerant as working fluid and the leakage of refrigerant would cause non-negligible damages to environment such as ozone layer destruction and greenhouse effect. Therefore, Montreal protocol was cosigned in 1987, respectively, for restricting the usage of CFC, HCFCs and Halon. After that, Kyoto protocol was further cosigned in 1997 for restricting the usage of PFCs, HFCs and SF₆. Thus, it is implied that to develop a heat pump system without using non-natural working fluid is unavoidable in the future.

For this reason, Stirling heat pump is concerned in this study because it uses natural gases as working fluid, such as air, nitrogen and helium, and it is environmentally friendly. In practical, Stirling heat pump is an inverse application of Stirling engine, which can pump heat from low-temperature reservoir to high-temperature reservoir by giving mechanical work. The working principle of Stirling heat pump is plotted in Figure 1. From process (A) to process (B), piston stays at the bottom dead point (BDP) and displacer moves from bottom dead point to top dead point (TDP). In the meantime, working fluid flows through regenerator and withdraws heat which is stored from process (C) to process (D). After that, working fluid is preheated and its temperature is increased. Now, from process (B) to process (C), displacer stays at TDP and piston moves to TDP. In this process, most of working fluid is located in compression space and heat

rejecter, which is compressed and rejects heat to high-temperature reservoir. In next step, piston stays at TDP and displacer moves to BDP as shown in process (C) to process (D). Working fluid is moved from compression space to expansion space, which passes through regenerator and store heat to regenerator at the same time. After that, most of working fluid is precooled and moved to expansion space and heat absorber. Finally, displacer stays at BDP and piston moves to BDP. In the meantime, working fluid is expanded and absorbs heat from low-temperature reservoir. By repeating above processes, then heat can be pumped from low-temperature reservoir to high-temperature reservoir continuously. On the other hand, the thermodynamic cycle of Stirling heat pump is an inverse Stirling cycle; therefore, the coefficient of performance (COP) of Stirling heat pump is equal to Carnot COP. Thus, Stirling heat pump with perfect regeneration has higher COP than other kinds of heat pump in theory. In addition, Stirling heat pump works as a closed gas cycle. Therefore, there is no exhausting problem and can be applied in some extreme environment, for example, high-altitude area, deep water and factory. Due to these advantages, Stirling heat pump has taken attention in the past decades. For example, Rix¹ developed a 3.7-kW alpha-type Stirling heat pump with Ross yoke mechanism and compared with theoretical predictions. Yagyu and coauthors^{2,3} tested and simulated a three-cylinder Stirling heat pump, which is driven by the engine and assisted by its exhaust heat. In the development of theory, Lucia and Gervino⁴ analyzed the thermo-economic properties of irreversible Stirling heat pump by using entropy generation. Tyagi and coauthors⁵ performed thermo-economic optimization and parametric study for an irreversible Stirling heat pump cycle with

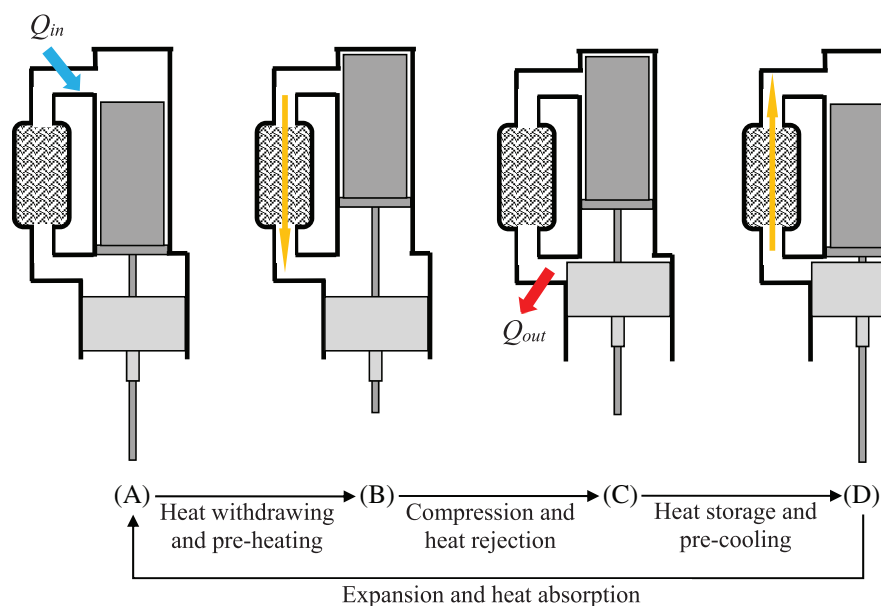
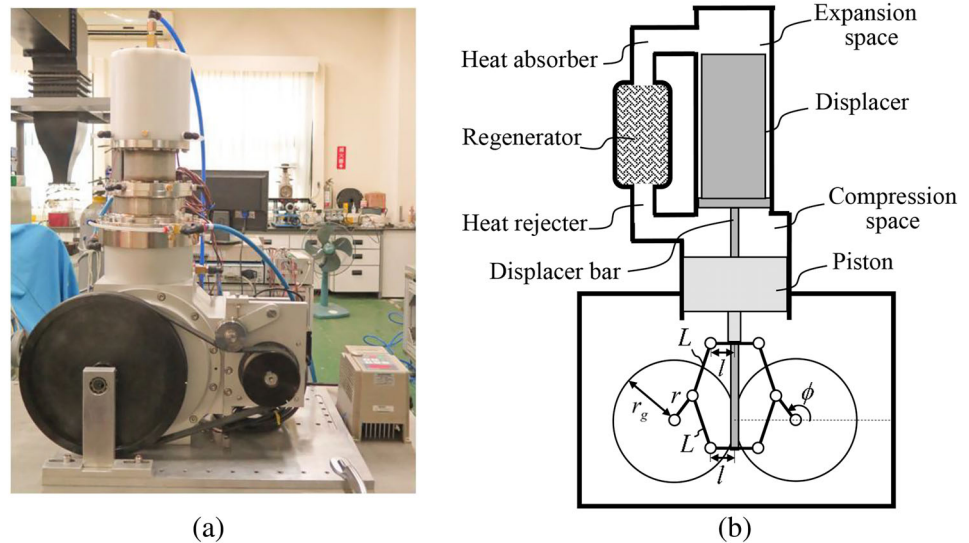


FIGURE 1 Working processes of the beta-type Stirling heat pump [Colour figure can be viewed at wileyonlinelibrary.com]

FIGURE 2 Photograph and schematic diagram of prototype Stirling heat pump [Colour figure can be viewed at wileyonlinelibrary.com]



the help of finite time thermodynamics method. Gadelkareem and coauthors⁶ applied Stirling cooler/heat pump on drinking water dispenser to produce hot and cold water simultaneously. A theoretical model was also proposed for simulating its performance.

In this study, a prototype beta-type Stirling heat pump for water heater is developed and tested. The photograph and schematic diagram are shown in Figure 2A,B, respectively. In this heat pump, there are two moving parts in the cylinder that are displacer and piston. The displacer is used to move working fluid between expansion space and compression space, and the piston is used to compress or expand working fluid in working space. Heat absorber is used to absorb heat from low-temperature reservoir. On the contrary, heat rejecter is used to reject heat to high-temperature reservoir. Regenerator is a heat exchanger for inner heat withdrawing and storage. In the present Stirling heat pump, a rhombic drive mechanism is adopted to drive the displacer and piston. In history, rhombic drive mechanism is first developed by Philips company in 1953.⁷ Because of the symmetry in Rhombic drive mechanism, the side friction force caused by driving mechanism can be eliminated and to decrease the mechanical loss effectively. In addition, rhombic drive mechanism is a well-known driving mechanism in Stirling machines, which has been applied widely in Stirling machines after 60 years of development. For example, Duan et al⁸ and Aksoy et al^{9,10} developed a beta-type Stirling engine with rhombic drive mechanism and performed a series of experimental investigation for their engine. Yang and coauthors^{11,12} proposed a modified non-ideal adiabatic model and combined it with dynamic model to simulate the performance and transient behavior of a beta-type Stirling engine with rhombic drive mechanism. Cheng and coauthors¹³ applied rhombic

drive mechanism on a Stirling cooler, which cooler can reach 93 K no-loading temperature at 1000 rpm with 3.039×10^5 Pa helium. Due to its reliability, a rhombic drive mechanism is adopted in this study.

In addition, for the purpose of designing a prototype Stirling heat pump, a numerical model is also established for predicting its performance. In this study, a modified non-ideal adiabatic model¹¹ is adopted as the thermodynamic model for simulating the variation of thermal properties of working fluid in working space, such as temperature, mass and pressure. It is different from Stirling engine that the temperature of the wall boundary of Stirling heat pump depends on operating conditions. Therefore, the energy equation of the wall boundary is also solved simultaneously in the present model. As a result, the transient variation of the thermal properties of working fluid and wall boundary in each working space can be predicted. Then, the performance of present heat pump, such as heating power, heating temperature and COP at transient state and steady state can be determined. Furthermore, the present model can be applied in parametric analysis and design optimization for improving the performance of present heat pump in the future. In order to validate the present model, a prototype of 1-kW class beta-type Stirling heat pump with rhombic drive mechanism is manufactured and tested in parallel. The experimental results are also compared with the simulation results in this study.

2 | NUMERICAL MODEL

2.1 | Volume variation

In this study, a beta-type Stirling heat pump with rhombic drive mechanism is designed and built. The schematic

diagram of the prototype Stirling heat pump is illustrated in Figure 2B. In general, the working space is divided into five main portions that are expansion space, heat absorber, regenerator, heat rejecter and compression space. These working spaces will be represented by subscripts e , a , r , h and c , respectively, in the following sections. In addition, there are two moving parts in this heat pump, namely, displacer and piston which are represented by subscripts d and p , respectively. Parameters r , r_g , l and L are the length of the linkages of rhombic drive mechanism. And the crank angle is represented by ϕ .

According to the geometric characteristics, the displacement function, top dead point and bottom dead point of displacer and piston can be presented in terms of ϕ , r , r_g , l and L . The detail derivation of the displacement function for rhombic drive mechanism can refer to Reference 14. Based on these results, the volume variations in expansion space and compression space caused by displacer motion and piston motion can be derived as below,

$$V_e = V_{e,s}(\chi_e + f_d), \quad (1)$$

$$V_c = V_{e,s}[\chi_c + (r_p - 1)f_p + (1 - r_b)f_{dp}] \quad \text{for } r_a \geq 1, \quad (2)$$

where $r_p = A_p/A_d$ and $r_b = A_b/A_d$ are the cross sectional area ratio, in which A_b , A_d and A_p are the cross sectional area of displacer bar, displacer and piston, respectively. The dead volume ratio in expansion space and compression space are represented by $\chi_e = V_{e, \text{dead}}/V_{e,s}$ and $\chi_c = V_{c, \text{dead}}/V_{e,s}$, respectively, where $V_{e, \text{dead}}$ and $V_{c, \text{dead}}$ are the dead volume in expansion space and compression space; functions f_d , f_p and f_{dp} are the displacement functions which are expressed in function of crank angle ϕ as below,

$$f_d = \left[-\sqrt{(e-1)^2 - \varepsilon^2} - \sin\phi + \sqrt{e^2 - (\varepsilon + \cos\phi)^2} \right] / \left[\sqrt{(e+1)^2 - \varepsilon^2} - \sqrt{(e-1)^2 - \varepsilon^2} \right], \quad (3)$$

$$f_p = \left[\sqrt{(e+1)^2 - \varepsilon^2} - \sin\phi - \sqrt{e^2 - (\varepsilon + \cos\phi)^2} \right] / \left[\sqrt{(e+1)^2 - \varepsilon^2} - \sqrt{(e-1)^2 - \varepsilon^2} \right], \quad (4)$$

$$f_{dp} = 2 \left[e - \sqrt{e^2 - (\varepsilon + \cos\phi)^2} \right] / \left[\sqrt{(e+1)^2 - \varepsilon^2} - \sqrt{(e-1)^2 - \varepsilon^2} \right], \quad (5)$$

where $e = L/r$ and $\varepsilon = (r_g - l)/r$ are the dimensionless parameters of rhombic drive mechanism. It can be found that the volume variation in compression space is not only caused by the piston motion but also by the displacer motion and their cross sectional area ratio. As a result, the volume variations in expansion space and compression space for a given design parameters r_p , r_b , e and ε can be determined. In this study, the design parameters for the prototype heat pump are $r_p = 1$, $r_b = 0.02$, $e = 2.5$ and $\varepsilon = 1.1$.

2.2 | Mass, pressure and temperature of working fluid

In this study, a modified non-ideal adiabatic model¹¹ is adopted as the thermodynamic model for simulating the thermal properties of working fluid in working spaces. The schematic diagram of present model is illustrated in Figure 3. As shown in Figure 3, working space is mainly divided into expansion space, heat absorber, regenerator, heat rejecter and compression space. To consider the effects of temperature gradient in heat absorber, regenerator and heat rejecter, these working spaces are further divided into 11 elements, respectively. In addition, every working space connected with its corresponding wall boundary.

The pressure and mass variations of working fluid in each working space caused by volume variations in expansion space and compression space can be evaluated first by using ideal adiabatic model¹⁵ as

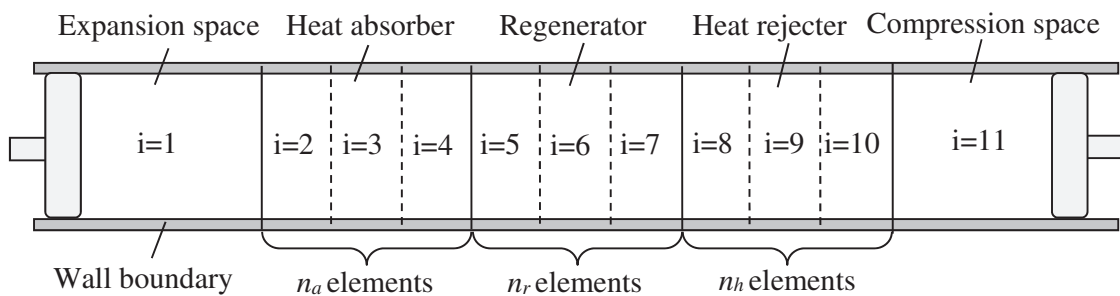


FIGURE 3 Schematic diagram of thermodynamic model

$$\frac{dp}{dt} = -\gamma p \left(\frac{1}{T_{1,2}} \frac{dV_1}{dt} + \frac{1}{T_{10,11}} \frac{dV_{11}}{dt} \right) / \left(\frac{V_1}{T_{1,2}} + \gamma \sum_{i=2}^{10} \frac{V_i}{T_i} + \frac{V_{11}}{T_{10,11}} \right), \quad (6)$$

where $V_1 = V_e$ and $V_{11} = V_c$. In Equation (6), p is the mean pressure of working fluid; V_i and T_i are the volume and temperature of working fluid, respectively, in i element; $T_{1,2}$ and $T_{10,11}$ are the temperature of working fluid which flow through the interface between two elements; $\gamma = c_{p,f}/c_{v,f}$ is the ratio of specific heats. The volume variations dV_1/dt and dV_{11}/dt can be obtained with the help of Equations (1) and (2), respectively. After dp/dt is obtained, then the corresponding mass variation in each working space caused by pressure variation can be calculated by

$$\frac{dm_1}{dt} = \frac{1}{RT_{1,2}} \left(p \frac{dV_1}{dt} + \frac{V_1 dp}{\gamma} \right), \quad (7)$$

$$\frac{dm_i}{dt} = \frac{V_i}{RT_i} \frac{dp}{dt} \quad \text{for } i = 2 \sim 10, \quad (8)$$

$$\frac{dm_{11}}{dt} = \frac{1}{RT_{10,11}} \left(p \frac{dV_{11}}{dt} + \frac{V_{11} dp}{\gamma} \right). \quad (9)$$

After that, the mass flow rate which flow through the interface between two elements can be obtained by

$$\dot{m}_{1,2} = -\frac{dm_1}{dt}, \quad (10)$$

$$\dot{m}_{i,i+1} = \dot{m}_{i-1,i} - \frac{dm_i}{dt} \quad \text{for } i = 2-9, \quad (11)$$

$$\dot{m}_{10,11} = \frac{dm_{11}}{dt}. \quad (12)$$

In which, $\dot{m}_{i,i+1} \geq 0$ means that the working fluid in each working space flows from element i to element $i+1$. On the contrary, $\dot{m}_{i,i+1} < 0$ means the working fluid flows in reverse direction. Then the energy equations for the temperature variation of working fluid T_i in each working space can be derived with the help of the first law of thermodynamics as below.

Expansion space:

$$\frac{dm_1 c_{v,f} T_1}{dt} = \frac{T_{w,1} - T_1}{R_1} - c_{p,f} \dot{m}_{1,2} T_{1,2} - p \frac{dV_1}{dt} + \frac{T_{11} - T_1}{R_{\text{loss}}}. \quad (13)$$

Heat absorber, regenerator and heat rejecter:

$$\frac{dm_i c_{v,f} T_i}{dt} = \frac{T_{w,i} - T_i}{R_i} + c_{p,f} (\dot{m}_{i-1,i} T_{i-1,i} - \dot{m}_{i,i+1} T_{i,i+1}) \quad \text{for } i = 2 \sim 10. \quad (14)$$

Compression space:

$$\frac{dm_{11} c_{v,f} T_{11}}{dt} = \frac{T_{w,11} - T_{11}}{R_{11}} + c_{p,f} \dot{m}_{10,11} T_{10,11} - p \frac{dV_{11}}{dt} - \frac{T_{11} - T_1}{R_{\text{loss}}}, \quad (15)$$

where $c_{v,f}$ and $c_{p,f}$ are the constant volume and constant pressure specific heat of working fluid. In this study, $c_{v,f}$ and $c_{p,f}$ are assumed as constant. The terms on the left hand side of Equations (13) to (15) are the internal energy variations of working fluid in each working space, which are dependent on the mass and temperature of working fluid. The first terms on the right hand side of Equations (13) to (15) are the heat transfer between working fluid and wall boundary, which is represented in terms of temperature difference and thermal resistance R_i . It is noted that the wall boundary in regenerator includes regenerator matrix and the cylinder which contains regenerator matrix. The value of R_i is dependent on the geometric design of flow channel, the thermal properties and flow state of working fluid, which can be calculated with the help of Nusselt number correlations.^{16,17} For the prototype heat pump, the cross sectional area of heat absorber and heat rejecter is circular and rectangular, respectively. The regenerator is composed of wire screen mesh. The corresponding Nusselt number correlations used in this study are

$$\text{Nu}_i = 3.657 \quad \text{for } i = 2 \sim 4, \quad (16)$$

$$\text{Nu}_i = 0.68 \text{Pr}^{0.33} \text{Re}^{0.6} \quad \text{for } i = 5 \sim 7, \quad (17)$$

$$\text{Nu}_i = 7.541 (1 - 2.61\alpha + 4.97\alpha^2 - 5.119\alpha^3 + 2.702\alpha^4 - 0.548\alpha^5) \quad \text{for } i = 8 \sim 10, \quad (18)$$

where Re and Pr are the Reynolds number and Prandtl number of mass flow rate, respectively, and α is the aspect ratio of rectangular duct.

The second terms on the right hand side of Equations (13) to (15) are the enthalpy exchanges between each element, which is dependent on the direction of mass flow rate $\dot{m}_{i-1,i}$ and the temperature of corresponding mass flow rate $T_{i-1,i}$. The third terms on the right hand side of Equations (13) and (15) are the expansion work and compression work caused by volume

variation. The last terms on the right hand side of Equations (13) and (15) are the total heat loss in working space, which may be caused by conduction loss due to working fluid, shuttle loss, pumping loss, etc.¹³ In general, total heat loss can be represented in terms of the temperature difference between expansion space and compression space and the total thermal resistance due to these heat losses. However, it can be found that the order of total heat loss is around 10^{-1} to 10^0 W, which can be neglected in a 1-kW class Stirling heat pump.

2.3 | Temperature of wall boundary

For a given temperature of wall boundary $T_{w,i}$, the temperature of working fluid T_i can be obtained by solving above Equations (13) to (15). However, in practice, the temperature of wall boundary $T_{w,i}$ in a real Stirling heat pump is unknown as it is operated, which have to be solved with the energy equations for $T_{w,i}$ simultaneously. The energy equations for $T_{w,i}$ can also be derived with the help of the first law of thermodynamics as below.

Expansion space:

$$m_{w,1}c_{v,w,1}\frac{dT_{w,1}}{dt} = \frac{T_1 - T_{w,1}}{R_1} + \frac{T_{sur,1} - T_{w,1}}{R_{sur,1}} - \frac{T_{w,1} - T_{w,2}}{R_{w,1,2}}. \quad (19)$$

Heat absorber, regenerator and heat rejecter:

$$m_{w,i}c_{v,w,i}\frac{dT_{w,i}}{dt} = \frac{T_i - T_{w,i}}{R_i} + \frac{T_{sur,i} - T_{w,i}}{R_{sur,i}} + \frac{T_{w,i-1} - T_{w,i}}{R_{w,i-1,i}} - \frac{T_{w,i} - T_{w,i+1}}{R_{w,i,i+1}} \text{ for } i = 2 \sim 10. \quad (20)$$

Compression space:

$$m_{w,11}c_{v,w,11}\frac{dT_{w,11}}{dt} = \frac{T_{11} - T_{w,11}}{R_{11}} + \frac{T_{sur,11} - T_{w,11}}{R_{sur,11}} + \frac{T_{w,10} - T_{w,11}}{R_{w,10,11}}, \quad (21)$$

where $m_{w,i}$ and $c_{v,w,i}$ on the left hand side of Equations (19) to (21) are the mass and specific heat of each wall boundary element, respectively. It is different from m_i that $m_{w,i}$ is constant with respect to time. However, the value of $m_{w,i}$ and $c_{v,w,i}$ may be different with respect to space due to the difference of material of each component. The first terms on the right hand side of Equations (19) to (21) are the heat exchanged between working fluid and wall boundary in each working space, which terms also appear on the right hand side of Equations (13) to (15) but with an opposed sign. The second terms on the right hand side of

Equations (19) to (21) are the heat exchanged between wall boundary and surrounding ambient of each working space, which also represented in terms of temperature difference and thermal resistance $R_{sur,i}$. In which, the temperature of surrounding ambient $T_{sur,i}$ is given, which may be constant or varied depends on the operating conditions. $R_{sur,i}$ may be contributed by conduction, convection or radiation, which is dependent on the design and application of Stirling heat pump. In this study, the regenerator is isolated from surrounding ambient; therefore, the second term on the right hand side of Equation (20) is neglected, namely, let $R_{sur,i} \rightarrow \infty$ as $i = 5-7$. The last terms on the right hand side of Equations (19) to (21) are the heat exchanged between wall elements due to conduction, which are also represented in terms of temperature difference and thermal resistance $R_{w,i-1,i}$. The value of $R_{w,i-1,i}$ is dependent on the geometric design, material and assemble method at the interface between different elements.

2.4 | Temperature of water in water jacket

In order to measure the heat absorbed and rejected by the heat pump, the water with constant mass flow rate and inlet temperature is adopted to flow through heat absorber and heat rejecter. The heat absorbed and rejected by the heat pump can be found by calculating the difference of internal energy of the water between inlet and outlet. The temperature variation of water in the water jacket of heat absorber can be predicted by solving

$$m_{a,wat}c_{v,wat}\frac{dT_{a,wat}}{dt} = \sum_{i=2}^4 \frac{T_{w,i} - T_{sur,i}}{R_{sur,i}} + \dot{m}_{a,wat}c_{v,wat}(T_{wat,in} - T_{a,wat,out}), \quad (22)$$

where $m_{a,wat}$ and $c_{v,wat}$ are the mass and specific heat of water in the water jacket of heat absorber, respectively, which values are assumed as constant here. The temperature of water in the water jacket of heat absorber is represented by $T_{a,wat}$, which value varies with time. The first term on the right hand side of Equation (22) presents the heat exchanged between the water in water jacket and the wall boundary of cylinder, which term also appears in Equations (19) to (21). Therefore, $T_{sur,i} = T_{a,wat}$ here. The second term on the right hand side of Equation (22) is the heat given by flow water. In which, the mass flow rate of water which pass through the water jacket of heat absorber is represented by $\dot{m}_{a,wat}$, which is assumed as a constant in this study. The temperature of input water is $T_{wat,in}$, which value is assumed as constant here. In addition, it is also assumed that the temperature of water which flow

out of the water jacket is equal to the temperature of water in water jacket, namely, $T_{a, \text{wat}, \text{out}} = T_{a, \text{wat}}$. Similarly, the temperature variation of water in the water jacket of heat rejecter can be predicted by solving

$$m_{h, \text{wat}} c_{v, \text{wat}} \frac{dT_{h, \text{wat}}}{dt} = \sum_{i=8}^{10} \frac{T_{w, i} - T_{\text{sur}, i}}{R_{\text{sur}, i}} + \dot{m}_{h, \text{wat}} c_{v, \text{wat}} (T_{\text{wat}, \text{in}} - T_{h, \text{wat}, \text{out}}), \quad (23)$$

where $m_{h, \text{wat}}$ and $\dot{m}_{h, \text{wat}}$ are the mass and mass flow rate of water in the water jacket of heat absorber, respectively. $T_{\text{sur}, i} = T_{h, \text{wat}} = T_{h, \text{wat}, \text{out}}$ here. It can be found obviously that T_i , $T_{w, i}$, $T_{a, \text{wat}}$ and $T_{h, \text{wat}}$ are coupled with each other by Equations (13) to (23), which have to be solved simultaneously.

After obtaining the temperature of working fluid and wall boundary in each elements, the pressure of working fluid in working space can be updated by

$$p = mR / \sum_{i=1}^{11} \frac{V_i}{T_i}. \quad (24)$$

It is noted that the denominator of Equation (24) is a summation of V_i/T_i in each working space. This formula can be obtained by assuming the temperature in each working space and the pressure in whole working space equilibrium instantaneously, which namely Schmidt assumptions.¹⁴ The pressure of working fluid in each working space can be further corrected by considering the effect of pressure drop, which can be estimated by using friction factor.^{16–18} After that, the mass of working fluid in each working space can be calculated by using ideal-gas equation of state,

$$m_i = pV_i / RT_i. \quad (25)$$

By repeating above processes, then the thermal properties of working fluid and the temperature of wall boundary in each element at every time step can be obtained.

2.5 | Performance of Stirling heat pump

After all the thermal properties are obtained, the performance of present heat pump can be calculated. The indicated power of every thermal cycle can be calculated by

$$\dot{W}_I = \frac{\omega}{2\pi} \oint p dV_e + \frac{\omega}{2\pi} \oint p dV_c. \quad (26)$$

The total heat input and output from the heat absorber and heat rejecter can be evaluated by

$$\dot{Q}_{\text{in}} = \frac{\omega}{2\pi} \oint \dot{m}_{a, \text{wat}} c_{v, \text{wat}} (T_{\text{wat}, \text{in}} - T_{a, \text{wat}, \text{out}}) dt, \quad (27)$$

$$\dot{Q}_{\text{out}} = \frac{\omega}{2\pi} \oint \dot{m}_{h, \text{wat}} c_{v, \text{wat}} (T_{\text{wat}, \text{in}} - T_{h, \text{wat}, \text{out}}) dt. \quad (28)$$

Finally, the coefficient of performance of the present heat pump can be obtained,

$$\text{COP} = |\dot{Q}_{\text{out}}| / (\dot{W}_I + \dot{W}_{\text{loss}}), \quad (29)$$

where $\dot{W}_{\text{loss}}$ is mechanical losses due to friction contributed by seals, piston rings, bearing and driving mechanism. For the present prototype heat pump, the value of $\dot{W}_{\text{loss}}$ under different operating speed can be calculated by following formula:

$$\dot{W}_{\text{loss}} = C_1 \omega^2 + C_2 \omega, \quad (30)$$

where $C_1 = 0.0159$ and $C_2 = 1.0174$. In this study, the value of $\dot{W}_{\text{loss}}$ under different operating speed can be obtained by measuring the operating torque given by motor as heat pump is at no-loading status which means that there is no pressure variation in the working space as the heat pump is driven by the motor. Then the coefficients in Equation (30) can be further determined by curve fitting the experimental data.

3 | EXPERIMENTAL APPARATUS

In this study, a prototype Stirling heat pump is manufactured and tested for confirming the present model.

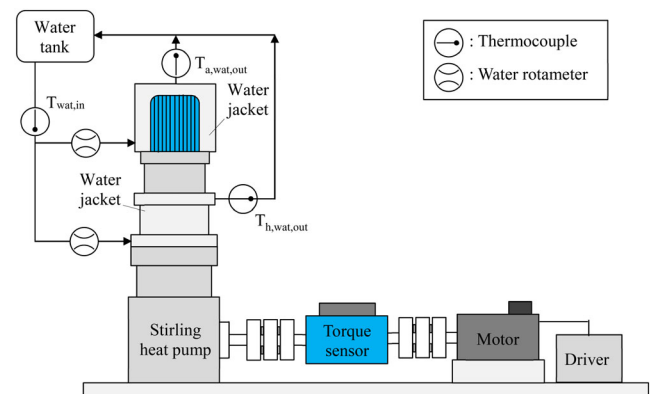


FIGURE 4 Schematic diagram of experimental apparatus [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Uncertainty analysis for measured variables

Variable	Typical value x	Uncertainty ^a δx	Relative uncertainty $\delta x/x$ (%)
T_{wat}	15–45°C	0.2°C	0.7
$\dot{m}_{\text{wat}}$	1 LPM	0.04 LPM	4.0
τ	2.5–7.5 N · m	0.03 N · m	0.6
ω	500–1000 rpm	15 rpm	2.0
COP	1–1.5		4.6 ^b

^aAll estimated with 95% confidence.

^b $\delta \text{COP}/\text{COP} = [(\delta T_{\text{wat}}/T_{\text{wat}})^2 + (\delta \dot{m}_{\text{wat}}/\dot{m}_{\text{wat}})^2 + (\delta \tau/\tau)^2 + (\delta \omega/\omega)^2]^{1/2}$.

The schematic diagram of experimental apparatus for testing the performance of the prototype Stirling heat pump is plotted as shown in Figure 4. In order to measure the rotating speed and torque input of the heat pump, a KISTLER torque sensor is adopted here which the measuring range of torque is 0 to 50 N · m and the maximum rotation speed is 12 000 rpm. A 3-kW brushless DC motor is used to drive the heat pump, which maximum rotating speed can reach 3000 rpm and can be controlled by a motor driver. Then the mechanical power input and mechanical losses under different operating speed can be measured by this equipment. The heat absorbed and rejected by the heat pump can be obtained by using Equations (22) and (23). In this study, the mass flow rate of water which flow through heat absorber and heat rejecter is measured by using individual water rotameter, respectively. The uncertainty of water rotameter is within $\pm 4\%$. The temperature of the water at the inlet and outlet of heat absorber and heat rejecter is measured by K-type thermocouple with $\pm 0.7\%$ uncertainty, which data is recorded by a data logger. With the help of this test platform, the temperature, heating power, COP and mechanical loss of prototype Stirling heat pump under different operating conditions can be measured.

In this study, uncertainty analysis of experimental results was performed by using the method presented by Moffat.¹⁹ The uncertainties of the measured quantities are all estimated with 95% confidence (Table 1). The experimental results reveal that the relative uncertainties of the rotation speed and torque input are around 2.0% and 0.6%, respectively. The relative uncertainty of COP is 4.6% by using the root sum-square method proposed in Reference 19.

4 | RESULTS AND DISCUSSION

The basic operating parameters of the prototype heat pump are listed in Table 2, which are also used in the present model. All the temperature of working fluid and

TABLE 2 Parameters of baseline case

Working fluid	Helium
Operating speed	1000 rpm
Charged pressure	5 bar
Ambient temperature (T_{sur})	298 K
Bore size of displacer and piston	70 mm
Stroke of displacer and piston	69 mm
Displacer swept volume ($V_{e, s}$)	266 c.c.
Compression ratio	1.51
Thermal resistance ($R_1 = R_n$)	0.01 K/W
Thermal resistance ($R_{\text{sur}, 2-4}$)	0.05 K/W
Thermal resistance ($R_{\text{sur}, 8-10}$)	5 K/W
Mass flow rate of water ($\dot{m}_{a, \text{wat}} = \dot{m}_{h, \text{wat}}$)	1 LPM

wall boundary at initial state is equal to the temperature of ambient surrounding.

Figure 5 shows the variation of heat input $\dot{Q}_{\text{in}}$, heat output $\dot{Q}_{\text{out}}$, indicated power $\dot{W}_I$ and COP with respect to time. At initial state, all the temperature of working fluid and wall boundary is equal to the temperature of ambient surrounding. Therefore, the value of $\dot{Q}_{\text{in}}$, $\dot{Q}_{\text{out}}$ and COP are equal to 0. After the heat pump is started, it can be found that heat absorber starts to absorb heat and pumps it to heat rejecter. The heat pump approach to steady state after about 10 minutes operating. In this period, the value of $\dot{W}_I$ increases from 510 to 605 W. In the meantime the value of $\dot{Q}_{\text{in}}$ increases from 0 to 577 W and the value of $\dot{Q}_{\text{out}}$ also increases from 0 to 1074 W. As a result, the value of COP increases from 0 to 1.775. It can be found that the terminal value of $\dot{Q}_{\text{out}}$ is less than the sum of $|\dot{W}_I + \dot{Q}_{\text{in}}|$ about 108 W, which difference is mainly caused by conduction loss.

Figure 6 shows the variation of cycle average temperature of working fluid and wall boundary in working space i . In which $i = 1, 3, 6, 9$ and 11 are represented expansion space, the middle of heat absorber, the middle of regenerator, the middle of heat rejecter and compression space, respectively. It can be found from Figure 6 that the temperature departs from 25°C after the heat pump is started, and then approaches to a new steady state in about 10 minutes. In low-temperature regions, the temperature of working fluid is lower than the temperature of wall boundary, namely, $T_1 < T_{w, 1}$ and $T_3 < T_{w, 3}$. As a result, the heat is absorbed from ambient surrounding to working fluid. The lowest temperature of wall boundary can reach 10°C. On the contrary, in high-temperature regions, the temperature of working fluid is higher than the temperature of wall boundary, namely, $T_9 > T_{w, 9}$ and $T_{11} > T_{w, 11}$. Therefore, the heat is rejected from working fluid to ambient surrounding in these regions. The highest temperature of wall

FIGURE 5 The variation of heat input, heat output, indicated power and COP with respect to time

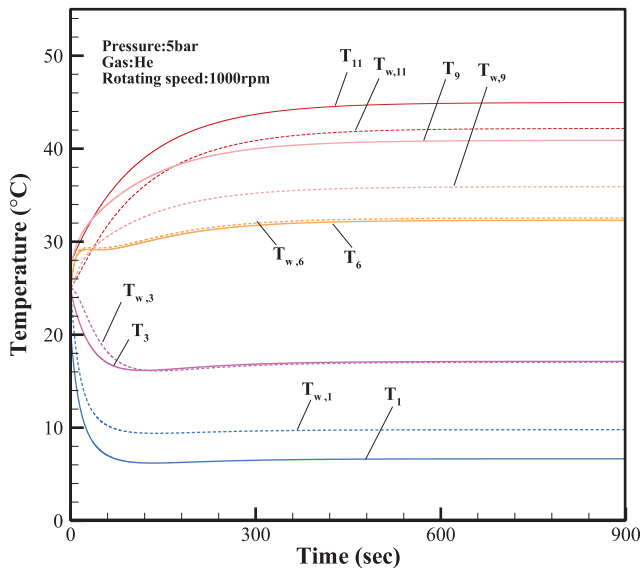
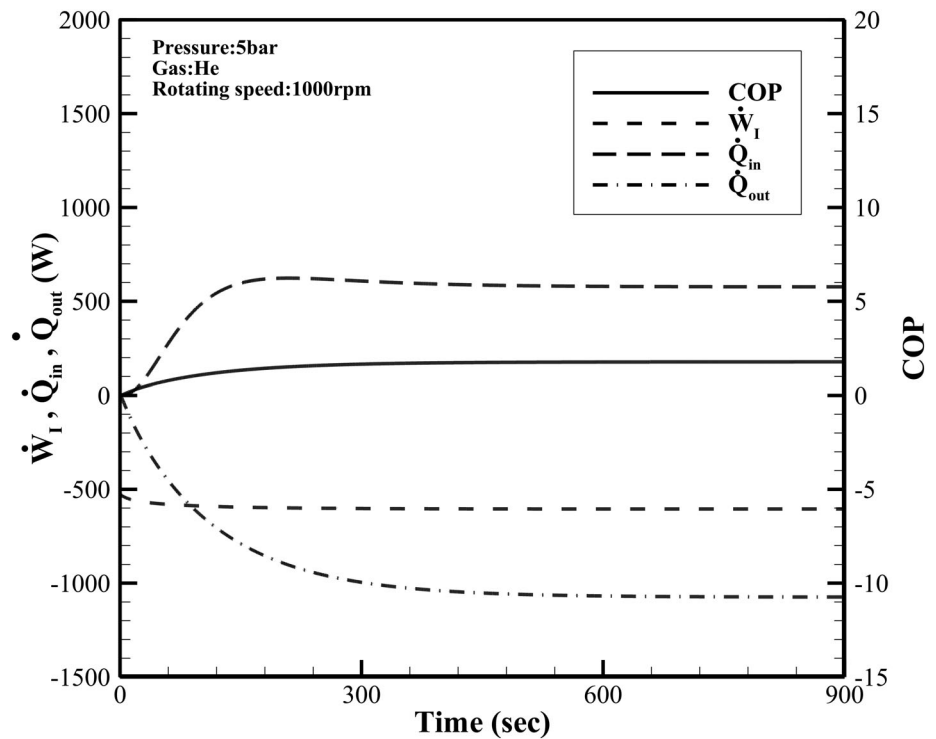


FIGURE 6 The variation of cycle average temperature of working fluid and wall boundary in working space $i = 1, 3, 6, 9$ and 11 [Colour figure can be viewed at wileyonlinelibrary.com]

boundary can reach 42°C . Figures 5 and 6 also reveal that the performance of prototype heat pump at transient and steady states can be predicted by the present model. By repeating above processes, then the performance of the heat pump under different design parameters and operating conditions can be predicted. Furthermore, it can be applied in parametric analysis and improving the design of prototype heat pump.

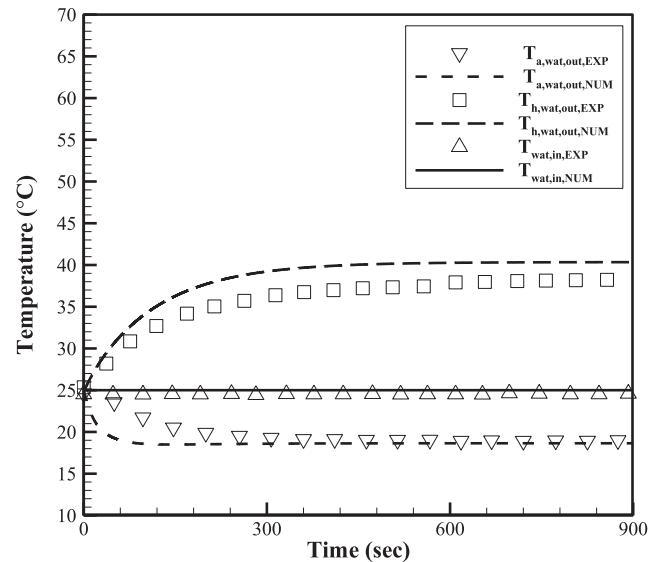


FIGURE 7 The temperature variation of water at the inlet and outlet of water jacket with respect to time

In order to validate the present numerical model, a comparison between experimental and simulation results was conducted. Figure 7 shows the temperature variation of water at the inlet and outlet of water jacket with respect to time after the heat pump starts running. In this case, the temperature of input water is about 25°C and with 1 LPM volume flow rate. In order to compare with simulation results, the value of $T_{wat, in}$ and $\dot{m}_{wat}$ in the present model is also set as 25°C and

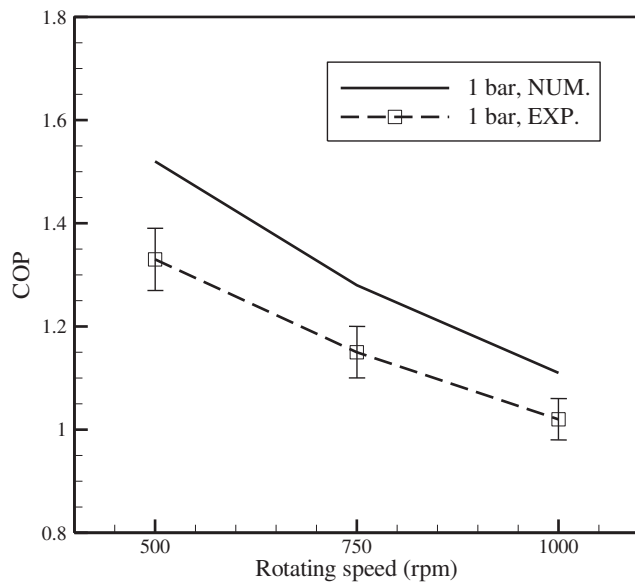


FIGURE 8 The comparison of COP between numerical and experimental results under 500, 750 and 1000 rpm with 1 bar helium

1 LPM. The experiment results show that the temperature of output water at the outlet of heat rejecter decreases from 25°C to about 19°C, and the temperature of output water at the outlet of heat absorber increases from 25°C to about 38°C after 15 minutes operating. The corresponding heat absorbed and rejected by the heat pump is about 417 and 904 W, respectively. On the other hand, the simulation results predict that the temperature of output water at the outlet of heat rejecter and heat absorber can approach to 18°C and 40°C after 15 minutes operating, respectively. The difference between experiment results and theoretical predictions is around 5%. It can be found that the simulation results agree with the experimental results well.

Figure 8 shows the comparison of COP between numerical and experimental results under 500, 750 and 1000 rpm with 1 bar helium. The experimental results indicate that the COP of prototype heat pump is 1.33, 1.15 and 1.02 at 500, 750 and 1000 rpm, respectively. The corresponding value of COP obtained from simulation results are 1.52, 1.28 and 1.11. The differences between experimental results and theoretical predictions are 14%, 11% and 9%. The results reveal that the error is higher at lower operating speed because the speed fluctuation is higher. It can be also found that the value of COP decreases with increasing operating speed monotonously, which is mainly caused by the imperfect heat exchange in working spaces in high-speed region and the mechanical losses as shown in Equation (30). Therefore, how to enhance the performance of heat exchanger in heat absorber and heat

rejecter and to decrease mechanical losses will be an important work in the future.

5 | CONCLUSIONS

In this paper, a numerical model for predicting the performance of a Stirling heat pump is proposed. To consider the terminal temperature of each component in Stirling heat pump is unknown, the effect of heat exchange between working fluid and wall boundary in each working space is taken into account. The transient variation of the thermal properties of the working fluid and wall boundary in each working space can be predicted by using present model as well as the transient performance variation. The present model can be further applied in parametric analysis and improving the design of prototype heat pump. In order to validate the present model, a 1-kW class prototype Stirling heat pump is built and a series of experimental measurement is also conducted and compared. The main findings in this study are summarized as follows:

1. Simulation results reveal that present Stirling heat pump begins to pump heat from heat absorber to heat rejecter after it started. The values of heat input and heat output increase from 0 to 577 W and 1074 W, respectively, in 10 minutes. The corresponding COP increases from 0 to 1.775. The results also indicate that the heat loss due to the conduction between heat absorber and heat rejecter causes a serious reduction in performance.
2. The results show that the temperature of compression space, heat rejecter and regenerator increase as the heat pump starts running, and the highest wall temperature can reach to 42°C. In the meantime, the temperature of expansion space and heat absorber decrease, and the lowest wall temperature can reach to 10°C.
3. The experimental results show that the temperature of the water at the outlet of heat rejecter can be cooled to 19°C and water at the outlet of heat absorber can be heated to 38°C under 1 LPM water flow rate with 5 bar helium at 1000 rpm. The corresponding cooling power and heating power are 417 and 904 W, respectively.
4. The results show that COP decreases with increasing operating speed, which is caused by imperfect heat exchange and mechanical losses. The maximum difference between experimental results and theoretical predictions is less than 14%. The results also reveal that the error is higher at lower operating speed because the speed fluctuation is higher.

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NOMENCLATURE

A	cross sectional area (m^2)
COP	coefficient of performance
C_1	coefficient in Equation (30) ($\text{N}\cdot\text{m}\cdot\text{s}$)
C_2	coefficient in Equation (30) ($\text{N}\cdot\text{m}$)
c_v	constant-volume specific heat ($\text{J}/\text{kg}\cdot\text{K}$)
c_p	constant-pressure specific heat ($\text{J}/\text{kg}\cdot\text{K}$)
e	L/r
f_d, f_p, f_{dp}	displacement function
L, l	lengths of links (m)
m	mass (kg)
$\dot{m}$	mass flow rate (kg/s)
Nu	Nusselt number
p	pressure (Pa)
Pr	Prandtl number
$\dot{Q}$	heat (W)
R	gas constant ($\text{J}/\text{kg}\cdot\text{K}$)
Re	Reynolds number
R	thermal resistance (K/W)
r	radius (m)
r_a, r_b	area ratio
t	time (s)
T	temperature (K)
V	volume (m^3)
$V_{e, s}$	volume swept by the displacer (m^3)
$\dot{W}_I$	indicated power (W)
$\dot{W}_{\text{loss}}$	mechanical losses (W)

GREEK SYMBOLS

α	aspect ratio of rectangular duct
χ	dead volume ratio
ε	$(r_g - 1)/r$
ϕ	crank angle (rad)
γ	specific heat ratio (c_p/c_v)
τ	torque ($\text{N}\cdot\text{m}$)
ω	angular velocity (rad/s)

SUBSCRIPTS

a	heat absorber
b	displacer bar
c	compression space
d	displacer
dead	dead volume
e	expansion space
f	working fluid
g	gear
h	heat rejecter
i	index of control volume
in	input

loss	loss
out	output
p	piston
r	regenerator
sur	ambient surrounding
w	wall
wat	water

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